

PREDICTIVE MODELING FOR COURSE DEMAND AND REVENUE FORECASTING ON EDUPRO

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PROJECT OVERVIEW

The EduPro Analytics Dashboard is an interactive Streamlit-based web application developed to analyze online course performance, predict course demand, and forecast revenue for an educational platform. The project integrates data analytics, machine learning, and visualization to provide actionable business intelligence across 19 courses, 3,081 learners, and \$819,294 in total platform revenue.

19 Total Courses	\$819,294 Total Revenue	3,081 Total Enrollments	2.89 / 5 Avg Course Rating
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METHODOLOGY & TECH STACK

- A structured 6-phase workflow was followed:
- Data collection from Excel sheets (Users, Teachers, Courses, Transactions)
 - Preprocessing & feature engineering (price bands, rating tiers, duration buckets)
 - Exploratory Data Analysis with Plotly & Streamlit visuals
 - ML modeling using Random Forest Regressor (Scikit-learn)
 - Visualization dashboard with KPI cards & interactive filters
 - Deployment as a Streamlit web application

Libraries: Python, Pandas, NumPy, Scikit-learn, Plotly, Streamlit

KEY FINDINGS

- Business (~180k) and Project Management (~170k) are the top revenue categories
- Artificial Intelligence leads growing demand at 15.2% of enrollments
- Project Management holds the highest enrollment share at 16.2%
- Course Price is the dominant predictor of revenue (importance ≈ 1.0)
- Finance (~28k) and Web Development (~10k) are the lowest earners
- Average rating of 2.89/5 highlights a course quality improvement opportunity
- Paid courses significantly outperform free courses in revenue contribution

MODEL PERFORMANCE

Metric	Value
Algorithm	Random Forest Regressor
R ² Score	0.965 (explains 96.5% of revenue variance)
Mean Absolute Error (MAE)	\$3,199.85
Top Predictive Feature	Course Price (importance ≈ 1.0)
Secondary Feature	Price Band (High category)

STRATEGIC RECOMMENDATIONS

- Expand Business and Project Management course portfolios to capitalize on strong demand and revenue performance.
- Invest heavily in AI/ML course content — 15.2% enrollment share reflects rising market demand.
- Redesign Finance and Web Development courses with updated content, targeted promotions, and revised pricing.
- Address average rating of 2.89/5 through instructor support, content audits, and learner feedback loops.

- Implement premium pricing strategy: course price is the single strongest revenue driver.
- Explore deep learning models and real-time cloud integration for improved forecast accuracy.